

# Confluence Modulo and Undecidability of Cut-Elimination\*

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## Abstract

Proofs in a logic are usually considered (at least) up to cut-elimination. This is the case in linear logic, that also has another equivalence relation on proofs: rule commutation. We prove these two notions coincide in second-order linear logic and many of its fragments. Moreover, we show that equality up to rule commutation is undecidable in propositional linear logic—and so is equality up to cut-elimination.

## 1 Introduction

A standard result in linear logic is the admissibility of the *cut*-rule, through a rewriting procedure called *cut-elimination* that turns a derivation (*a.k.a.* proof tree) into another derivation with no *cut*-rule. Through the Curry-Howard correspondence, cut-elimination is linked to  $\beta$ -reduction of  $\lambda$ -calculus, and it is sensible to consider derivations up to cut-elimination, exactly as  $\lambda$ -terms are considered up to  $\beta$ -reduction. This is often done in semantics, especially in denotational models like coherent spaces [Gir87] or Seely categories [See89]. To consider objects up to a rewriting procedure, the royal road is having *strong normalization* and *confluence*. They yield together the *unique normal form* property, giving a *canonical* representative for each equivalence class. Alas, cut-elimination is not that well-behaved: it enjoys neither strong normalization nor confluence.

**No strong normalization.** The *cut* – *cut* step of cut-elimination can be repeated *ad nauseam*:

$$\frac{\frac{\frac{\vdash A^\perp, B^\perp, \Gamma}{\vdash B^\perp, \Gamma, \Delta} \text{ (cut)} \quad \vdash A, \Delta}{\vdash B, \Sigma} \text{ (cut)} \longrightarrow \frac{\frac{\frac{\vdash A^\perp, B^\perp, \Gamma}{\vdash A^\perp, \Gamma, \Sigma} \text{ (cut)} \quad \vdash B, \Sigma}{\vdash A, \Delta} \text{ (cut)} \text{ (cut)}$$

**No confluence.** A derivation can be rewritten to different cut-free forms:

$$\frac{\frac{\frac{\vdash C^\perp, C}{\vdash C^\perp, C} \text{ (ax)} \quad \frac{\frac{\vdash A^\perp, A}{\vdash A^\perp, A} \text{ (ax)} \quad \frac{\vdash B^\perp, B}{\vdash B^\perp, B} \text{ (ax)}}{\vdash A^\perp, A \otimes B^\perp, B} \text{ (}\otimes\text{)} \longleftarrow + \frac{\frac{\frac{\vdash C^\perp, C}{\vdash C^\perp, C} \text{ (ax)} \quad \frac{\vdash A^\perp, A}{\vdash A^\perp, A} \text{ (ax)}}{\vdash C^\perp, C \otimes A^\perp, A} \text{ (}\otimes\text{)} \quad \frac{\frac{\vdash A^\perp, A}{\vdash A^\perp, A} \text{ (ax)} \quad \frac{\vdash B^\perp, B}{\vdash B^\perp, B} \text{ (ax)}}{\vdash A^\perp, A \otimes B^\perp, B} \text{ (}\otimes\text{)} \xrightarrow{+} \frac{\frac{\frac{\vdash C^\perp, C}{\vdash C^\perp, C} \text{ (ax)} \quad \frac{\vdash A^\perp, A}{\vdash A^\perp, A} \text{ (ax)}}{\vdash C^\perp, C \otimes A^\perp, A} \text{ (}\otimes\text{)} \quad \frac{\vdash B^\perp, B}{\vdash B^\perp, B} \text{ (ax)}}{\vdash C^\perp, C \otimes A^\perp, A \otimes B^\perp, B} \text{ (}\otimes\text{)} \text{ (cut)}$$

Consequently, the literature on cut-elimination in sequent calculus is not as developed as one may wish: there is not much more than weak normalization, *i.e.* admissibility of the *cut*-rule—see for instance [Gir95; Oka99]. Nonetheless, such a result has been proved for many variations of linear logic, see *e.g.* [EP16; Acc22; AMM25; BS25; BS26]. Naturally, not everything about cut-elimination revolves around normalization and confluence, and there are many works where cut-elimination is central, *e.g.* around session types [DeY+12] or interpolation [Sau25]. Let us mention cut-elimination has been well-studied for *proof-nets* [Gir96], the graphical syntax of linear logic. This framework is closely linked to sequent calculus, but there strong normalization and confluence (usually) hold, leading to a very rich literature on the subject for many fragments and extensions of linear logic, in *e.g.* [Tor03; LM08; DG99; Acc13; Gue+24; Pag09; Tra09; PT17; PT10].

In sequent calculus, since strong normalization and confluence fail we have to consider weaker properties. The failure of strong normalization is only due to the *cut* – *cut* step: provided this step does not occur infinitely many times, one has strong normalization. Besides, cut-elimination is confluent *up to rule commutation*: normal forms differ only by the order in which their rules are applied—see our example above. Derivations are

\*Related to an extended draft [Di26].

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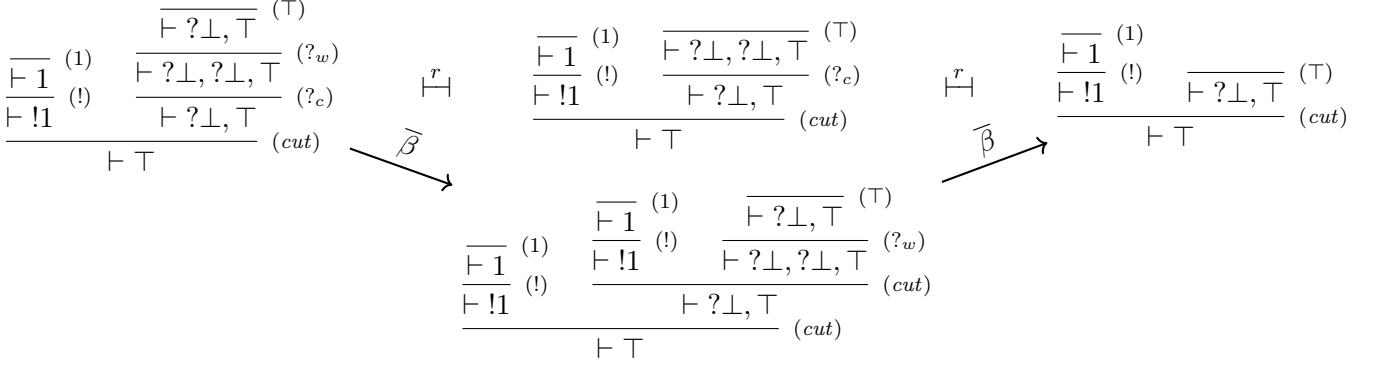


Figure 1: Counter-example to the strong normalization of  $\xrightarrow{\beta} \cdot \vdash^*$

often considered up to rule commutation, which is viewed as “bureaucracy” as all orders of rules are adequate but we cannot apply several rules at once. This explains why cut-elimination behaves better in proof-nets, that quotient derivations exactly by rule commutations [HG16]. This key result of confluence up to has so far only been proved for propositional multiplicative-additive linear logic [CP05, Theorem 5.1; DL25, Theorem 4.2], but was expected for the whole logic. We prove it, showing equality up to cut-elimination is simply rule commutations (between cut-free derivations). Equality up to rule commutation is simpler to study, and we show it is *undecidable* in propositional linear logic—and so is equality up to cut-elimination.

## 2 Second-order Linear Logic

Our framework is the standard sequent calculus of linear logic [Gir87] *with units and second-order quantifiers*. It is equipped with **cut-elimination**  $\xrightarrow{\beta}$  and **rule commutation**  $\vdash^*$ . Some points of attention:

- we separate steps of cut-elimination as  $\xrightarrow{\beta} = \xrightarrow{\bar{\beta}} \cup \vdash^c$  with the *cut – cut* commutation  $\vdash^c$  on one side, and all other rules  $\xrightarrow{\bar{\beta}}$  on the other side (to obtain strong normalization);
- rule commutations  $\vdash^*$  are the usual ones, except we assume there is no *cut*-rule above the rules that commute, and we do *not* have commutations with *!*-rules (not needed for confluence up to):

$$\frac{\frac{\vdash A, ?B, ?B, ?\Gamma}{\vdash !A, ?B, ?B, ?\Gamma} (!)}{\vdash !A, ?B, ?\Gamma} (?_c) \quad \frac{\vdash A, ?B, ?B, ?\Gamma}{\vdash A, ?B, ?\Gamma} (?_c) \quad \frac{\vdash A, ?\Gamma}{\vdash !A, ?\Gamma} (!) \quad \frac{\vdash A, ?\Gamma}{\vdash A, ?B, ?\Gamma} (?_w) \quad \frac{\vdash A, ?\Gamma}{\vdash !A, ?B, ?\Gamma} (?_w) \quad \frac{\vdash A, ?B, ?\Gamma}{\vdash !A, ?B, ?\Gamma} (!)$$

## 3 Rewriting theory modulo an equivalence relation

We assume known standard definitions and notations from rewriting theory [Ter03]: normal forms,  $\xrightarrow{\alpha^+}$  is the transitive closure of  $\xrightarrow{\alpha}$ , etc. We will prove cut-elimination is Church-Rosser modulo rule commutation.

**Definition 1** (Church-Rosser modulo). Let  $\sim$  and  $\rightarrow$  be relations on a set with  $\sim$  an equivalence relation. We say  $\rightarrow$  is **Church-Rosser modulo**  $\sim$  when  $(\rightarrow \cup \leftarrow \cup \sim)^* \subseteq \rightarrow^* \cdot \sim \cdot \leftarrow^*$ .

The literature contains various results for proving Church-Rosser modulo, *e.g.* [JK84; Ohl98; AT12; Fel24]. Some are generalizations of Newman’s Lemma [Ter03, Theorem 1.2.1] to prove Church-Rosser modulo instead of confluence. Unfortunately, most results do not apply to our case, in particular none of those from [Ter03, Section 14.3], with two due to Huet [Hue80, Lemmas 2.7, 2.8] and one to van Oostrom [Oos94, Proposition 2.5.3]. First, closing diagrams with as hypotheses  $a \vdash^* b$  is too complex, preventing from applying *e.g.* [Hue80, Lemma 2.7; Oos94, Proposition 2.5.3]. Second, strong normalization of  $\xrightarrow{\beta} \cdot \vdash^*$  is false—see Figure 1 for a counter-example—preventing from applying *e.g.* [Hue80, Lemma 2.8]. We use the following result, heavily inspired by a theorem from Aoto and Toyama [AT12, Theorem 2.2]. Intuitively, its item (ii) generalizes local confluence (*a.k.a.* property  $\alpha$ ) and its item (iii) local coherence (*a.k.a.* property  $\gamma$ ).

**Proposition 2.** Let  $\rightarrow$ ,  $\vdash$  and  $\rightsquigarrow$  be relations on a set  $A$  such that  $\vdash$  is symmetric and  $\rightsquigarrow \subseteq \vdash$ . Set  $\Rightarrow := \rightarrow \cup \rightsquigarrow$ . Suppose:

- (i)  $\rightarrow \cdot \rightsquigarrow^*$  is strongly normalizing;
- (ii)  $\leftarrow \cdot \rightarrow \subseteq \Rightarrow^* \cdot \bar{\vdash} \cdot * \leftarrow$ ;
- (iii) given  $a, b, c \in A$ , if  $a \vdash b \rightarrow c$  then  $a \rightarrow \cdot \Rightarrow^* \cdot \bar{\vdash} \cdot * \leftarrow \cdot \leftarrow b$ .

Then  $\rightarrow$  is Church-Rosser modulo  $\bar{\vdash}$ .

*Proof.* The relation  $(\rightsquigarrow^* \cdot \rightarrow \cdot \rightsquigarrow^*)^*$  is a well-founded strict partial order on  $A$ : transitivity is immediate, and irreflexivity and well-foundedness come from the strong normalization of  $\rightarrow \cdot \rightsquigarrow^*$ . For a sequence  $\rho$  of elements of  $A$  related by either  $\vdash$ ,  $\rightarrow$  or  $\leftarrow$ , pose the multiset  $\mathcal{M}(\rho) := \uplus_{a \vdash b \in \rho} \{a, a, b, b\} \uplus \uplus_{a \rightarrow b \in \rho} \{a\} \uplus \uplus_{a \leftarrow b \in \rho} \{b\}$ , equipped with a well-founded strict partial order  $<$  by the multiset extension of the one on  $A$ .

We proceed by induction on  $\mathcal{M}(\rho)$ . If  $\rho$  is of the shape  $\rightarrow^* \cdot \bar{\vdash} \cdot * \leftarrow$  then we are done. Otherwise,  $\rho$  contains a sub-sequence  $a \leftarrow b \rightarrow c$ , or  $a \vdash b \rightarrow c$ , or  $c \leftarrow b \vdash a$ . If  $\rho$  contains a sub-sequence  $a \leftarrow b \rightarrow c$ , we replace it with  $a \Rightarrow^* \cdot \bar{\vdash} \cdot * \leftarrow c$ ; the resulting sequence  $\rho'$  respects  $\mathcal{M}(\rho') < \mathcal{M}(\rho)$ , and we conclude by induction hypothesis. If  $\rho$  contains a sub-sequence  $a \vdash b \rightarrow c$  or  $c \leftarrow b \vdash a$ , we replace  $a \vdash b$  with  $a \rightarrow \cdot \Rightarrow^* \cdot \bar{\vdash} \cdot * \leftarrow b$ , yielding a new sequence  $\rho'$  with  $\mathcal{M}(\rho') < \mathcal{M}(\rho)$ , and conclude by induction.  $\square$

## 4 Church-Rosser modulo for cut-elimination

We prove cut-elimination  $\xrightarrow{\bar{\beta}}$  is Church-Rosser modulo rule commutations  $\bar{\vdash}^*$ , by showing  $\xrightarrow{\bar{\beta}}$  is Church-Rosser modulo  $(\bar{\vdash} \cup \vdash^c)^*$  through Proposition 2. So, we prove strong normalization and close some diagrams.

### 4.1 Strong normalization up to rule commutation

As shown on Figure 1,  $\xrightarrow{\bar{\beta}} \cdot \bar{\vdash}^*$  is not strongly normalizing due to commutations with  $\top$ . Ergo, we *orient* those: note  $\bar{\vdash}^{\top}$  the commutations without  $\top$ , and  $\xrightarrow{\top}$  those with  $\top$  oriented in the “erasing” direction, e.g.:

$$\frac{\frac{}{\vdash A, B, \top, \Gamma} (\top)}{\vdash A \wp B, \top, \Gamma} (\wp) \xrightarrow{\top} \frac{}{\vdash A \wp B, \top, \Gamma} (\top) \qquad \frac{\frac{}{\vdash \top, A[B/X], \Gamma} (\top)}{\vdash \top, \exists X A, \Gamma} (\exists) \xrightarrow{\top} \frac{}{\vdash \top, \exists X A, \Gamma} (\top)$$

Note  $\bar{\vdash}^r = \bar{\vdash}^{\top} \cup \xrightarrow{\top} \cup \xrightarrow{\top}$ . To apply Proposition 2, we need strong normalization of  $\xrightarrow{\bar{\beta}} \cdot (\bar{\vdash}^c \cup \bar{\vdash}^{\top} \cup \xrightarrow{\top})^*$ . This almost follows from a technical paper by Michele Pagani and Lorenzo Tortora de Falco [PT10] proving strong normalization of cut-elimination in proof-nets for second-order linear logic, using some non-standard proof-nets called *sliced pure structures*. The idea underlying our proof is a simple adaptation of [PT10]: there is a translation  $\mathcal{T}$  from derivations of sequent calculus to sliced pure structures that satisfies the following.

- Images of  $\mathcal{T}$  are easily shown to be strongly normalizing for a reduction procedure  $\longrightarrow$ .
- If  $\pi \xrightarrow{\bar{\beta}} \phi$  or  $\pi \xrightarrow{\top} \phi$  then  $\mathcal{T}(\pi) \xrightarrow{*} \mathcal{T}(\phi)$ .
- If  $\pi \vdash^c \phi$  or  $\pi \bar{\vdash}^{\top} \phi$  then  $\mathcal{T}(\pi) = \mathcal{T}(\phi)$ .

Strong normalization for derivations then easily follows by studying a possible infinite reduction. Nonetheless, this is not a corollary of [PT10] but of a generalization of it with a few more reduction rules (corresponding to  $\xrightarrow{\top}$ ) and a slightly modified definition of sliced pure structures (to have invariance by  $\top - \top$  commutation). Thence, we check that every following definition and statement in [PT10] still holds after these modifications.

**Theorem 3.** The relation  $\xrightarrow{\bar{\beta}} \cdot (\bar{\vdash}^c \cup \bar{\vdash}^{\top} \cup \xrightarrow{\top})^*$  is strongly normalizing.

Figure 2: A critical pair for the proof of Lemma 4

Figure 3: A critical pair for the proof of Lemma 6

## 4.2 Diagrams to close & Church-Rosser modulo

Items (ii) and (iii) of Proposition 2 with  $\rightarrow := \xrightarrow{\beta}$ ,  $\vdash := \vdash^c \cup \vdash^r$  and  $\rightsquigarrow := \vdash^c \cup \vdash^r \cup \xrightarrow{\tau}$  are given by:

**Lemma 4.** *If  $\pi_1 \xleftarrow{\beta} \pi \xrightarrow{\beta} \pi_2$ , then  $\pi_1 \vdash^c \cdot \xrightarrow{\beta^*} \cdot (\vdash^r \cup \vdash^c) \cdot \xleftarrow{\beta^*} \pi_2$ .*

**Lemma 5.** *If  $\pi_1 \vdash^r \pi_2 \xrightarrow{\beta} \pi_3$ , then  $\pi_1 \xrightarrow{\beta^+} \cdot \xrightarrow{\tau^*} \cdot (\vdash^r \cup \vdash^c) \cdot \xleftarrow{\tau^*} \cdot \xleftarrow{\beta^+} \pi_2$ .*

**Lemma 6.** *If  $\pi_1 \vdash^c \pi_2$ , then  $\pi_1 \xrightarrow{\beta^+} \cdot \vdash^c \cdot \xrightarrow{\beta^*} \cdot \xleftarrow{\beta^*} \cdot \vdash^c \cdot \xleftarrow{\beta^+} \pi_2$ .*

Proofs are case studies: a critical pair for Lemma 4 (resp. Lemma 6) is solved in Figure 2 (resp. Figure 3).

**Proposition 7.** *Cut-elimination  $\xrightarrow{\beta}$  is Church-Rosser modulo rule commutation  $\vdash^*$ .*

*Proof.* By Proposition 2, Theorem 3, and Lemmas 4 to 6,  $\xrightarrow{\beta}$  is Church-Rosser modulo  $\vdash^r \cup \vdash^c$ . Consider two derivations  $\pi$  and  $\phi$  such that  $\pi (\xrightarrow{\beta} \cup \xleftarrow{\beta} \cup \vdash^r)^* \phi$ . By Theorem 3, there exist cut-free derivations  $\pi'$  and  $\phi'$  such that  $\pi \xrightarrow{\beta^*} \pi'$  and  $\phi \xrightarrow{\beta^*} \phi'$ . Remark  $\pi' (\xrightarrow{\beta} \cup \xleftarrow{\beta} \cup \vdash^c \cup \vdash^r)^* \phi'$ . As  $\xrightarrow{\beta}$  is Church-Rosser modulo  $\vdash^r \cup \vdash^c$ , and  $\pi'$  and  $\phi'$  are  $\xrightarrow{\beta}$ -normal forms, it follows that  $\pi' (\vdash^c \cup \vdash^r)^* \phi'$ . But no  $\vdash^c$  step can be applied on cut-free derivations, and  $\vdash^r$  preserves cut-freeness:  $\pi' \vdash^* \phi'$ , concluding the proof.  $\square$

## 5 Equality up to cut-elimination / up to rule commutation

We get our main result: two derivations are equal up to cut-elimination if and only if their normal forms are related by rule commutations. The direct way is Proposition 7, and the converse is simple but tedious.

**Proposition 8** ( $\vdash^r \subseteq =_\beta$ ). *Given derivations  $\pi$  and  $\tau$ , if  $\pi \vdash^r \tau$  then  $\pi \xleftarrow{\bar{\beta}^+} \cdot \xrightarrow{\bar{\beta}^+} \tau$ .*

*Proof.* All cases are similar, we present only one (see it in Click  $\mathfrak{A}$  c $\otimes$ LLe $\perp$  [here](#)).

$$\frac{\overline{\vdash A \& B, \top, \Gamma}^{(\top)} \xleftarrow{\bar{\beta}^+} \overline{\vdash \top, 0}^{(\top)} \quad \frac{\overline{\vdash A, \top, \Gamma}^{(\top)} \quad \overline{\vdash B, \top, \Gamma}^{(\top)}}{\overline{\vdash A \& B, \top, \Gamma}^{(\&)}} \xrightarrow{\bar{\beta}^+} \frac{\overline{\vdash A, \top, \Gamma}^{(\top)} \quad \overline{\vdash B, \top, \Gamma}^{(\top)}}{\overline{\vdash A \& B, \top, \Gamma}^{(\&)}} \quad (cut)$$

□

**Theorem 9.** *Two derivations are equal up to cut-elimination  $\xrightarrow{\beta}$  if and only if any of their  $\xrightarrow{\beta}$ -normal forms are related by rule commutations  $\vdash^*$ .*

*Proof.* Consider derivations  $\pi$  and  $\phi$  with respective  $\xrightarrow{\beta}$ -normal forms  $\pi'$  and  $\phi'$ . Note  $\pi =_\beta \phi \iff \pi' =_\beta \phi'$ . If  $\pi' =_\beta \phi'$  then  $\pi' \vdash^* \phi'$  (Proposition 7 on cut-free derivations). If  $\pi' \vdash^* \phi'$  then  $\pi' =_\beta \phi'$  (Proposition 8). □

Our proof of Theorem 9 applies to *most* fragments of second-order linear logic: only Proposition 8 (and then Theorem 9) needs some hypotheses for the exponentials.

**Theorem 10.** *Two derivations are equal up to cut-elimination  $\xrightarrow{\beta}$  if and only if any of their  $\xrightarrow{\beta}$ -normal forms are related by rule commutations  $\vdash^*$ , in all fragments of linear logic such that:*

- if the  $?_c$ -rule belongs to the fragment, then the  $\mathfrak{A}$ - and  $\otimes$ -rules also belong to it;
- if the  $?_w$ -rule belongs to the fragment, then the  $\perp$ - and  $1$ -rules also belong to it.

## 6 Undecidability of equality up to rule commutation/cut-elimination

We now consider only propositional linear logic, *without quantifiers*. We examine three problems:

**Equality up to Cut:** Given two derivations, are they equal up to cut-elimination?

**Proof Equivalence:** Given two cut-free derivations, are they equal up to rule commutation?

**Provability:** Given a sequent, is it provable?

By Theorem 9, Equality up to Cut is undecidable if and only if Proof Equivalence is. We reduce to the later Provability, well-known to be undecidable [Lin95].

**Lemma 11.** *For any formula  $A$ ,  $\left( \frac{\overline{\vdash !A \otimes \top, \top}^{(\top)}}{\overline{\vdash !A \otimes \top, \top \oplus \top}^{(\oplus_1)}} \vdash^* \frac{\overline{\vdash !A \otimes \top, \top}^{(\top)}}{\overline{\vdash !A \otimes \top, \top \oplus \top}^{(\oplus_2)}} \right) \iff A \text{ is provable}$*

*Proof.* If  $A$  is provable, then we exhibit rule commutations linking these two derivations. If  $A$  is not provable, then we can compute the equivalence classes of both derivations. □

**Proposition 12.** *Proof Equivalence and Equality up to Cut are undecidable in propositional linear logic.*

*Proof.* Provability is undecidable [Lin95]. By Lemma 11, this entails undecidability of Proof Equivalence, which entails by Theorem 9 undecidability of Equality up to Cut. □

## 7 Conclusion

Cut-elimination in second-order linear logic is Church-Rosser modulo rule commutation, a result that was widely expected but not yet proved. Hence, equality up to cut-elimination coincides with rule commutations on cut-free derivations. This holds in most sub-systems of linear logic, *e.g.* multiplicative-exponential linear logic. Also, adding *e.g.* the *mix*-rules is easy. Furthermore, we can extend this in presence of other usual transformations of derivations, such as axiom-expansion.

We prove equality up to rule commutation is undecidable in propositional linear logic. This completely prevents any canonical notion of proof-nets for propositional linear logic, and adding the *mix*-rule does not solve the problem for our proof easily extends to that setting. This contrasts with the case of multiplicative linear logic where adding the *mix*-rule turns the problem from PSPACE-complete [HH16] to solvable by proof-nets—see [FR94] for the first definition of these proof-nets.

## References

- [Acc13] Beniamino Accattoli. “Linear Logic and Strong Normalization”. In: *Rewriting Techniques and Applications*. Ed. by Femke van Raamsdonk. Vol. 21. 2013. DOI: [10.4230/LIPIcs.RTA.2013.39](https://doi.org/10.4230/LIPIcs.RTA.2013.39).
- [Acc22] Beniamino Accattoli. “Exponentials as Substitutions and the Cost of Cut Elimination in Linear Logic”. In: *Proceedings of the 37th Annual ACM/IEEE Symposium on Logic in Computer Science (LICS)*. Haifa, Israel, 2022. DOI: [10.1145/3531130.3532445](https://doi.org/10.1145/3531130.3532445).
- [AMM25] Matteo Acclavio, Giulia Manara, and Fabrizio Montesi. “Formulas as Processes, Deadlock-Freedom as Choreographies”. In: *Programming Languages and Systems*. Ed. by Viktor Vafeiadis. Springer Nature Switzerland, 2025, pp. 23–55. DOI: [10.1007/978-3-031-91118-7\\_2](https://doi.org/10.1007/978-3-031-91118-7_2).
- [AT12] Takahito Aoto and Yoshihito Toyama. “A Reduction-Preserving Completion for Proving Confluence of Non-Terminating Term Rewriting Systems”. In: *Logical Methods in Computer Science* 8.1 (Mar. 2012), pp. 1–29. DOI: [10.2168/LMCS-8\(1:31\)2012](https://doi.org/10.2168/LMCS-8(1:31)2012).
- [BS25] Esaïe Bauer and Alexis Saurin. “On the cut-elimination of the modal  $\mu$ -calculus: Linear Logic to the rescue”. In: *Foundations of Software Science and Computation Structures*. Ed. by Parosh Aziz Abdulla and Delia Kesner. Springer, 2025, pp. 133–154. DOI: [10.1007/978-3-031-90897-2\\_7](https://doi.org/10.1007/978-3-031-90897-2_7).
- [BS26] Esaïe Bauer and Alexis Saurin. “A Uniform Cut-Elimination Theorem for Linear Logics with Fixed Points and Super Exponentials”. In: *34th EACSL Annual Conference on Computer Science Logic (CSL 2026)*. Ed. by S. Guerrini and B. König. 2026. DOI: [10.4230/LIPIcs.CSL.2026.17](https://doi.org/10.4230/LIPIcs.CSL.2026.17).
- [CP05] Robin Cockett and Craig Pastro. “A Language For Multiplicative-additive Linear Logic”. In: *Electronic Notes in Theoretical Computer Science* 122 (2005), pp. 23–65. DOI: [/10.1016/j.entcs.2004.06.049](https://doi.org/10.1016/j.entcs.2004.06.049).
- [DeY+12] Henry DeYoung, Luis Caires, Frank Pfenning, and Bernardo Toninho. “Cut Reduction in Linear Logic as Asynchronous Session-Typed Communication”. In: *Computer Science Logic*. Ed. by Patrick Cégielski and Arnaud Durand. Vol. 16. 2012. DOI: [10.4230/LIPIcs.CSL.2012.228](https://doi.org/10.4230/LIPIcs.CSL.2012.228).
- [DG99] Roberto Di Cosmo and Stefano Guerrini. “Strong Normalization of Proof Nets Modulo Structural Congruences”. In: *Rewriting Techniques and Applications*. Ed. by P. Narendran and M. Rusinowitch. Vol. 1631. Lecture Notes in Computer Science. 1999. DOI: [10.1007/3-540-48685-2\\_6](https://doi.org/10.1007/3-540-48685-2_6).
- [Di26] Rémi Di Guardia. “Confluence Modulo and Undecidability of Cut-Elimination in Linear Logic”. Draft. 2026. URL: [https://remidig.github.io/papers/confluence\\_cut\\_ll.pdf](https://remidig.github.io/papers/confluence_cut_ll.pdf).
- [DL25] Rémi Di Guardia and Olivier Laurent. “Type Isomorphisms for Multiplicative-Additive Linear Logic”. In: *Logical Methods in Computer Science* 16.4 (Nov. 2025). DOI: [10.46298/lmcs-21\(4:24\)2025](https://doi.org/10.46298/lmcs-21(4:24)2025).
- [EP16] Harley Eades and Valeria de Paiva. “Multiple Conclusion Linear Logic: Cut Elimination and More”. In: *Logical Foundations of Computer Science*. Ed. by Sergei Artemov and Anil Nerode. Springer International Publishing, 2016, pp. 90–105. DOI: [10.1007/978-3-319-27683-0\\_7](https://doi.org/10.1007/978-3-319-27683-0_7).
- [Fel24] Thiago Felicissimo. “Second-order Church-Rosser modulo, without normalization”. In: *Proceedings of the 13th International Workshop on Confluence - IWC 2024*. July 2024. URL: <https://hal.science/hal-04835978v1>.
- [FR94] Arnaud Fleury and Christian Retoré. “The Mix Rule”. In: *Mathematical Structures in Computer Science* 4.2 (1994), pp. 273–285. DOI: [10.1017/S0960129500000451](https://doi.org/10.1017/S0960129500000451).
- [Gir87] Jean-Yves Girard. “Linear logic”. In: *Theoretical Computer Science* 50 (1987), pp. 1–102. DOI: [10.1016/0304-3975\(87\)90045-4](https://doi.org/10.1016/0304-3975(87)90045-4).
- [Gir95] Jean-Yves Girard. “Linear logic: its syntax and semantics”. In: *Advances in Linear Logic*. Ed. by Jean-Yves Girard, Yves Lafont, and Laurent Regnier. Vol. 222. London Mathematical Society Lecture Note Series. Cambridge University Press, 1995, pp. 1–42.
- [Gir96] Jean-Yves Girard. “Proof-nets: the parallel syntax for proof-theory”. In: *Logic and Algebra*. Ed. by Aldo Ursini and Paolo Agliano. Vol. 180. Lecture Notes In Pure and Applied Mathematics. New York: Marcel Dekker, 1996, pp. 97–124. DOI: [10.1201/9780203748671-4](https://doi.org/10.1201/9780203748671-4).



- [GLR95] Jean-Yves Girard, Yves Lafont, and Laurent Regnier, eds. *Advances in Linear Logic*. Vol. 222. London Mathematical Society Lecture Note Series. Cambridge University Press, 1995.
- [Gue+24] Giulio Guerrieri, Giulia Manara, Lorenzo Tortora de Falco, and Lionel Vaux Auclair. “Confluence for Proof-Nets via Parallel Cut Elimination”. In: *Proceedings of 25th Conference on Logic for Programming, Artificial Intelligence and Reasoning*. Ed. by Nikolaj Bjørner, Marijn Heule, and Andrei Voronkov. Vol. 100. EPIc Series in Computing. 2024, pp. 464–483. DOI: [10.29007/vkfn](https://doi.org/10.29007/vkfn).
- [HG16] Dominic Hughes and Rob van Glabbeek. *MALL proof nets identify proofs modulo rule commutation*. 2016. URL: <https://arxiv.org/abs/1609.04693>.
- [HH16] Willem Heijltjes and Robin Houston. “Proof equivalence in MLL is PSPACE-complete”. In: *Logical Methods in Computer Science* 12.1 (2016). DOI: [10.2168/LMCS-12\(1:2\)2016](https://doi.org/10.2168/LMCS-12(1:2)2016).
- [Hue80] Gérard Huet. “Confluent Reductions: Abstract Properties and Applications to Term Rewriting Systems”. In: *Journal of the ACM* 27.4 (Oct. 1980), pp. 797–821. DOI: [10.1145/322217.322230](https://doi.org/10.1145/322217.322230).
- [JK84] Jean-Pierre Jouannaud and Helene Kirchner. “Completion of a set of rules modulo a set of equations”. In: *Proceedings of the 11th ACM SIGACT-SIGPLAN Symposium on Principles of Programming Languages*. POPL ’84. 1984, pp. 83–92. DOI: [10.1145/800017.800519](https://doi.org/10.1145/800017.800519).
- [Lin95] Patrick Lincoln. “Deciding provability of linear logic formulas”. In: *Advances in Linear Logic*. Ed. by Jean-Yves Girard, Yves Lafont, and Laurent Regnier. Vol. 222. London Mathematical Society Lecture Note Series. Cambridge University Press, 1995, pp. 109–122.
- [LM08] Olivier Laurent and Roberto Maieli. “Cut Elimination for Monomial MALL Proof Nets”. In: *Proceedings of the twenty-third annual symposium on Logic In Computer Science*. IEEE. IEEE Computer Society Press, June 2008, pp. 486–497.
- [Ohl98] Enno Ohlebusch. “Church-Rosser theorems for abstract reduction modulo an equivalence relation”. In: *Rewriting Techniques and Applications*. Ed. by Tobias Nipkow. Vol. 1379. Lecture Notes in Computer Science. Springer, 1998, pp. 17–31. ISBN: 978-3-540-69721-3.
- [Oka99] Mitsuhiro Okada. “Phase Semantic cut elimination and normalization proofs of first- and higher-order linear logic”. In: *Theoretical Computer Science* 227 (Sept. 1999), pp. 333–396.
- [Oos94] Vincent van Oostrom. “Confluence for Abstract and Higher-Order Rewriting”. Ph.D. thesis. Vrije Universiteit Amsterdam, 1994.
- [Pag09] Michele Pagani. “The Cut-Elimination Theorem for Differential Nets with Promotion”. In: *Typed Lambda Calculi and Applications ’09*. Ed. by Pierre-Louis Curien. Vol. 5608. Lecture Notes in Computer Science. Springer, July 2009, pp. 219–233.
- [PT10] Michele Pagani and Lorenzo Tortora de Falco. “Strong normalization property for second order linear logic”. In: *Theoretical Computer Science* 411.2 (2010). DOI: [10.1016/j.tcs.2009.07.053](https://doi.org/10.1016/j.tcs.2009.07.053).
- [PT17] Michele Pagani and Paolo Tranquilli. “The Conservation Theorem for Differential Nets”. In: *Mathematical Structures in Computer Science* 27.6 (2017). DOI: [10.1017/S0960129515000456](https://doi.org/10.1017/S0960129515000456).
- [Sau25] Alexis Saurin. “Interpolation as Cut-Introduction: On the Computational Content of Craig-Lyndon Interpolation”. In: *International Conference on Formal Structures for Computation and Deduction (FSCD)*. Ed. by Maribel Fernández. 2025. DOI: [10.4230/LIPIcs.FSCD.2025.32](https://doi.org/10.4230/LIPIcs.FSCD.2025.32).
- [See89] Robert Seely. “Linear logic,  $\star$ -autonomous categories and cofree coalgebras”. In: *Contemporary mathematics* 92 (1989).
- [Ter03] Terese. *Term Rewriting Systems*. Vol. 55. Cambridge tracts in theoretical computer science. Cambridge University Press, 2003.
- [Tor03] Lorenzo Tortora de Falco. “Additives of linear logic and normalization- Part I: a (restricted) Church-Rosser property”. In: *Theoretical Computer Science* 294.3 (Feb. 2003), pp. 489–524. DOI: [10.1016/S0304-3975\(01\)00176-1](https://doi.org/10.1016/S0304-3975(01)00176-1).
- [Tra09] Paolo Tranquilli. “Confluence of Pure Differential Nets with Promotion”. In: *Proceedings of the 23rd CSL International Conference and 18th EACSL Annual Conference on Computer Science Logic*. CSL’09/EACSL’09. Springer, 2009, pp. 500–514. DOI: [10.5555/1807662.1807701](https://doi.org/10.5555/1807662.1807701).